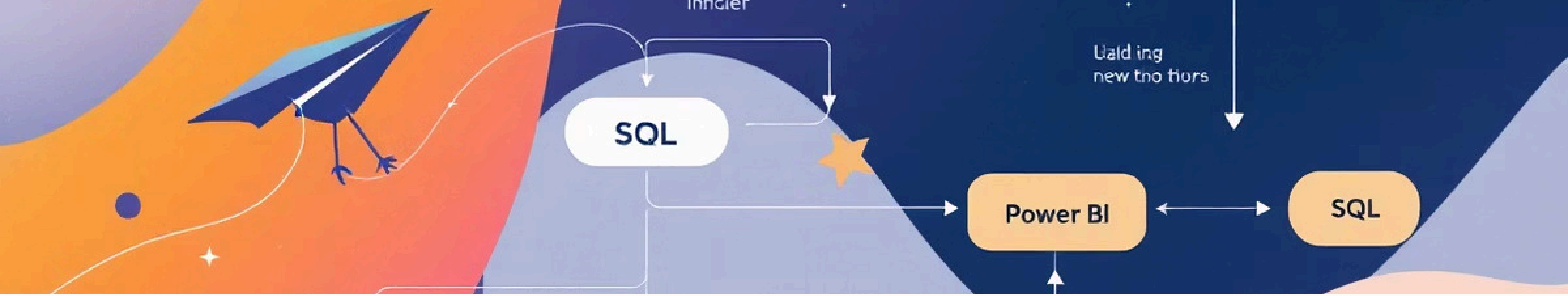
Customer Insights Dashboard

End-to-end data analytics report uncovering actionable insights into customer shopping behavior



Project Overview

Dataset Scope

3,900 customers analyzed

Total Revenue

$233,081 generated

Tools Used

SQL, Python, Power BI (DAX)

This project demonstrates an industry-aligned workflow integrating Python for data cleaning, SQL for exploratory analysis, and Power BI for interactive visualization with custom KPIs.

Revenue Analysis by Gender

SQL analysis revealed purchasing power distribution across demographics. Male customers generate higher total revenue compared to female customers, indicating stronger revenue contribution within this dataset.

Result Grid | Filter Rows:

gender	Net_Revenue
Male	157890
Female	75191

$233K

Total Revenue

Generated across all customers

Critical Finding: Subscription Gap

Key Insight: Analysis reveals that **none of the subscribed customers are female.** This indicates a significant gap in subscription adoption among female customers.

Gather Insights

Understand female customer preferences and barriers

Targeted Campaigns

Introduce new policies and subscription benefits

Attract & Retain

Focus on female audience engagement

	item_purchased	Net_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	[illegible]
	Handbag	3.78

Top-Rated Products

Average customer review ratings identify top-performing items, with footwear and accessories receiving highest satisfaction scores.

Gloves

Highest average rating

Sandals

Strong customer satisfaction

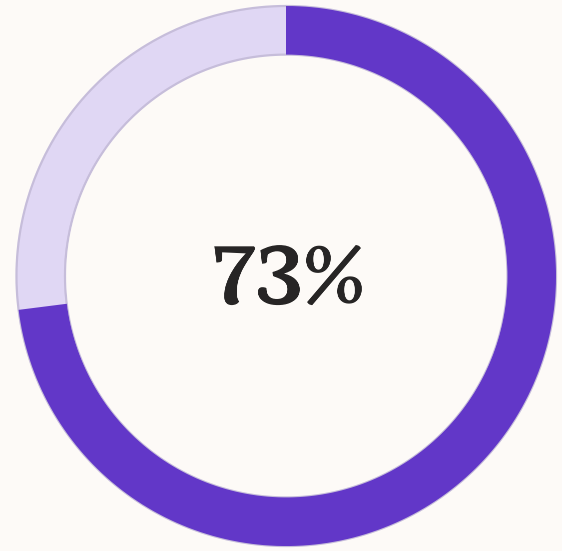
Boots

Top footwear category

Subscription Performance

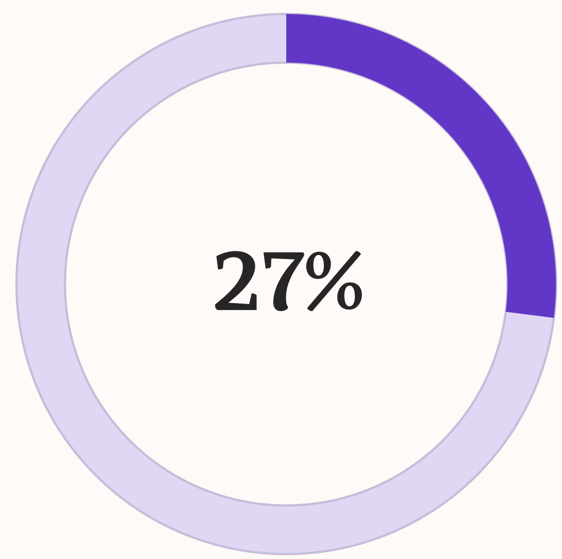
Subscriber vs. Non-Subscriber Metrics

- Non-subscribers: $170,436 total revenue (2,847 customers)
- Subscribers: $62,645 total revenue (1,053 customers)
- Average sales per customer comparable between groups (~$59)



Non-Subscribers

Larger revenue share



Subscribers

Growth opportunity

Customer Segmentation

SQL analysis using Common Table Expressions (CTE) segmented customers based on purchase history, revealing retention strength.

New Customers

1 previous purchase

Returning Customers

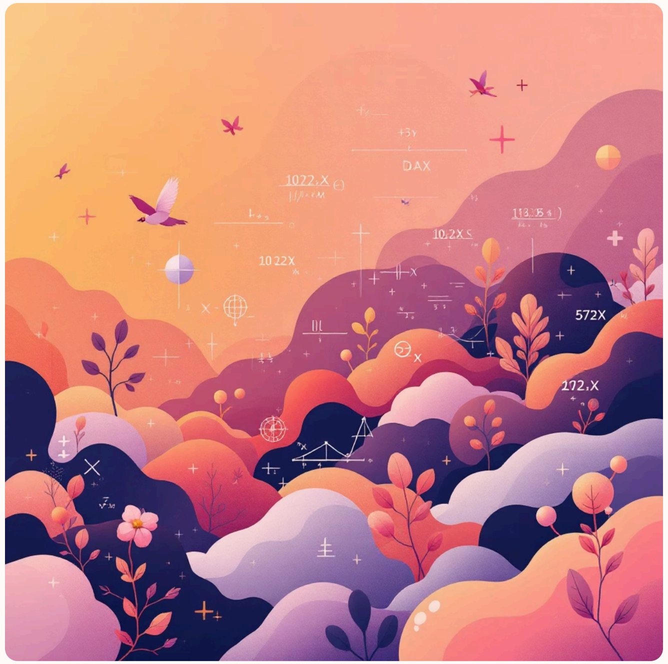
2-10 previous purchases

Loyal Customers

10+ purchases (majority segment)

Customer_segment	Number_of_Customers
Loyal	3116
Returning	701
New	83

Power BI Dashboard Implementation



DAX Measures

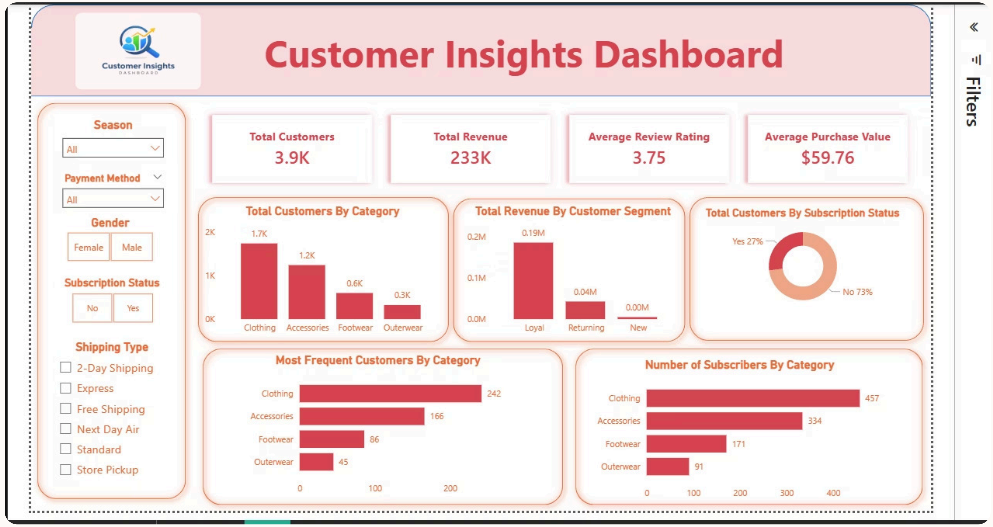
Custom measures for Total Customers, Average Transaction Value, and Average Rating.

SQL Validation

Cross-validated SQL results with DAX measures for accuracy.

Interactive Visuals

KPI cards, donut charts, and bar charts with dynamic filters.



Live Dashboard

This is the actual Power BI dashboard developed during the project. It showcases the interactive visualizations, KPI metrics, and dynamic filters discussed in the previous section, providing a real-time view of our data insights.

Key Performance Indicators

$233K

Total Revenue

Across all transactions

3,900

Total Customers

Analyzed in dataset

3.75

Average Rating

Customer satisfaction score

$59

Avg Transaction

Per customer purchase

Revenue breakdown by category shows strong performance across Clothing, Accessories, and Footwear segments.

Strategic Recommendations

Voice of Customer Survey

Launch targeted survey for female shoppers to understand subscription barriers and preferences

Ladies' Exclusive Tier

Develop tailored membership tier with benefits specifically designed for female customers

Customer Acquisition

Focus on new customer acquisition while maintaining strong loyal customer base

This project demonstrates strong proficiency in SQL-driven analysis, KPI development using DAX, and interactive dashboard creation—an industry-aligned, end-to-end data analytics workflow.